Statistical Analysis of the Palmer Penguins Dataset

DTS 204 – Statistical Computing, Inference and Modelling: Lab Assessment

ADEMOLA UTIBE AFOLABI
Email: a.afolabi3364@miva.edu.ng
Department: Computer Science
Matric Number: 2025/A/CSC/0096
Course: DTS 204
MIVA Open University
GitHub: github.com/CodewithDemmy/DTS204_Lab_Report_files
24 October 2026

Contents

1 Introduction and objectives **3**

2 Methods overview **3**

3 Task 1: Data exploration and preprocessing **4**

3.1 Loading the data and the first six rows 4

3.2 Missing values 5

3.3 Summary statistics 6

4 Task 2: Hypothesis testing **7**

4.1 Two-sample t-test: body mass of Adelie vs Gentoo penguins 7

4.2 One-sample proportion test: proportion of male penguins 9

5 Task 3: Correlation analysis **10**

5.1 Correlation matrix 10

5.2 Heatmap 11

5.3 Discussion of highly correlated variables 11

6 Task 4: Regression modelling **13**

6.1 Simple linear regression: body mass predicted by flipper length 13

6.2 Multiple linear regression 14

6.3 Comparing the two models 16

6.4 Checking the regression assumptions 17

7 Task 5: Visualization **18**

7.1 Scatter plot of body mass vs flipper length with the regression line 18

7.2 Boxplot of body mass across species 19

7.3 Histogram of body mass with a normal curve 20

8 Summary of findings in plain language **22**

9 Assumptions and limitations **22**

9.1 Assumptions behind the methods 22

9.2 Limitations 22

10 Conclusion **23**

11 Appendix: reproducibility **23**

1 Introduction and objectives

This report analyses the **Palmer Penguins** dataset, which records measurements of penguins observed on three islands (Biscoe, Dream and Torgersen) in the Palmer Archipelago, Antarctica. Three species are represented: Adelie, Chinstrap and Gentoo. For each bird the dataset gives its species, island, bill length, bill depth, flipper length, body mass and sex.

The analysis answers the following questions, one section per lab task:

1. **Data exploration (Task 1):** What does the data look like, is anything missing, and what are the basic summary statistics?
2. **Hypothesis testing (Task 2):** Do Adelie and Gentoo penguins differ in average body mass? Is the share of male penguins different from 50%?
3. **Correlation (Task 3):** Which body measurements move together?
4. **Regression (Task 4):** How well can body mass be predicted from flipper length, and does adding the two bill measurements help?
5. **Visualization (Task 5):** Scatter plot, boxplot and histogram of the key variables.
6. **Reporting (Task 6):** Plain-language interpretation of every finding, plus the assumptions and limitations of the analysis.

All code is shown in the report and is also supplied as a stand-alone script (`penguins_analysis.R`). Every number quoted in the text is calculated by R when the report is built, so the text always matches the output.

2 Methods overview

```r
library(palmerpenguins)
library(ggplot2)
library(dplyr)
library(knitr)

alpha <- 0.05   # significance level used for every test
species_cols <- c(Adelie = "#E69F00", Chinstrap = "#0072B2",
                  Gentoo = "#009E73")

# palmerpenguins:: is written out in full because recent versions of R
# also contain a different dataset called "penguins".
penguins <- palmerpenguins::penguins
```

The analysis was run in R version 4.6.1 (2026-06-24 ucrt). The general approach is:

Task	Method used	Why
1. Exploration	Head of the data, missing-value count, mean, median, range, SD	Understand the data before testing anything
2i. Body mass, Adelie vs Gentoo	Welch two-sample t-test (two-sided)	Compares two group means without assuming equal variances

Task	Method used	Why
2ii. Proportion of males	One-sample proportion test against 0.5	Tests a single proportion against a benchmark value
3. Correlation	Pearson correlation matrix, correlation tests, heatmap	Measures linear association between measurements
4. Regression	Simple and multiple linear regression (least squares)	Predicts body mass and compares models
5. Visualization	Scatter plot, boxplot, histogram (ggplot2)	Shows the patterns behind the numbers

All hypothesis tests use a significance level of $\alpha = 0.05$ and are two-sided. A result with $p < 0.05$ is called *statistically significant*.

3 Task 1: Data exploration and preprocessing

3.1 Loading the data and the first six rows

```
dim(penguins)
```

```
#> [1] 344   8
```

```
head_tbl <- as.data.frame(head(penguins, 6))
```

Table 1: First six rows of the penguins dataset (bill and flipper lengths in mm, body mass in g).

Species	Island	Bill len.	Bill depth	Flipper	Mass	Sex	Year
Adelie	Torgersen	39.1	18.7	181	3750	male	2007
Adelie	Torgersen	39.5	17.4	186	3800	female	2007
Adelie	Torgersen	40.3	18.0	195	3250	female	2007
Adelie	Torgersen	NA	NA	NA	NA	NA	2007
Adelie	Torgersen	36.7	19.3	193	3450	female	2007
Adelie	Torgersen	39.3	20.6	190	3650	male	2007

The dataset has 344 rows (one per penguin) and 8 columns. Besides the seven variables described in the lab brief, the package also includes a `year` column (the year of observation, 2007 to 2009), which is a label rather than a measurement and is therefore left out of the numerical summaries.

3.2 Missing values

```
missing_summary <- data.frame(
  variable    = names(penguins),
  n_missing   = as.integer(colSums(is.na(penguins))),
  pct_missing = round(100 * colMeans(is.na(penguins)), 2)
)
incomplete_rows <- penguins[!complete.cases(penguins), ]
n_incomplete <- nrow(incomplete_rows)
n_complete   <- sum(complete.cases(penguins))
n_meas       <- sum(!is.na(penguins$body_mass_g))
n_sex        <- sum(!is.na(penguins$sex))
```

Table 2: Missing values per variable.

Variable	Missing values	Percent missing
species	0	0.00
island	0	0.00
bill_length_mm	2	0.58
bill_depth_mm	2	0.58
flipper_length_mm	2	0.58
body_mass_g	2	0.58
sex	11	3.20
year	0	0.00

Of the 344 penguins, 333 rows are fully complete and 11 rows have at least one missing value:

- 2 penguins have **no measurements at all** (bill length, bill depth, flipper length and body mass are all missing).
- 11 penguins have an unrecorded **sex**. Two of them are the penguins above; the other 9 were measured but not sexed.

How the missing values are handled. The amount of missing data is very small (under 1% for the measurements and about 3% for sex), so it is unlikely to change any conclusion. I therefore use *analysis-specific deletion*: each test or model uses every penguin that has the variables it needs.

- Summary statistics, correlations, regressions and the t-test use the 342 penguins that have measurements.
- The proportion test uses the 333 penguins with a recorded sex.
- I do **not** fill in (impute) missing values. The two penguins with no measurements contain no information to impute from, and guessing the sex of the others would bias the proportion test, which is the very quantity being tested.

3.3 Summary statistics

```
num_vars <- c("bill_length_mm", "bill_depth_mm",
              "flipper_length_mm", "body_mass_g")
nums <- as.data.frame(penguins)[num_vars]

summary_stats <- data.frame(
  variable = num_vars,
  n      = sapply(nums, function(x) sum(!is.na(x))),
  mean   = sapply(nums, mean,   na.rm = TRUE),
  median = sapply(nums, median, na.rm = TRUE),
  min    = sapply(nums, min,    na.rm = TRUE),
  max    = sapply(nums, max,    na.rm = TRUE),
  sd     = sapply(nums, sd,     na.rm = TRUE),
  row.names = NULL
)
summary_stats$range <- summary_stats$max - summary_stats$min
summary_stats <- summary_stats[, c("variable", "n", "mean", "median",
                                   "min", "max", "range", "sd")]
```

Table 3: Summary statistics for the numerical variables.

Variable	n	Mean	Median	Min	Max	Range	SD
Bill length (mm)	342	43.92	44.45	32.1	59.6	27.5	5.46
Bill depth (mm)	342	17.15	17.30	13.1	21.5	8.4	1.97
Flipper length (mm)	342	200.92	197.00	172.0	231.0	59.0	14.06
Body mass (g)	342	4201.75	4050.00	2700.0	6300.0	3600.0	801.95

```
body_mass_by_species <- penguins %>%
  group_by(species) %>%
  summarise(n      = sum(!is.na(body_mass_g)),
            mean   = mean(body_mass_g,   na.rm = TRUE),
            median = median(body_mass_g, na.rm = TRUE),
            sd     = sd(body_mass_g,     na.rm = TRUE),
            min    = min(body_mass_g,    na.rm = TRUE),
            max    = max(body_mass_g,    na.rm = TRUE),
            .groups = "drop") %>%
  as.data.frame()
```

Table 4: Body mass by species.

Species	n	Mean (g)	Median (g)	SD (g)	Min (g)	Max (g)
Adelie	151	3700.7	3700	458.6	2850	4775
Chinstrap	68	3733.1	3700	384.3	2700	4800
Gentoo	123	5076.0	5000	504.1	3950	6300

In plain language. A typical penguin in this dataset weighs about 4,202 g (roughly 4.2 kg), with a flipper about 201 mm long and a bill about 43.9 mm long and 17.2 mm deep. Body mass varies a lot: the lightest bird weighs 2,700 g and the heaviest 6,300 g, a range of 3,600 g, with a standard deviation of about 802 g. For all four variables the mean and median are close, which suggests no extreme outliers. Much of the variation in body mass comes from differences between species: Gentoo penguins average 5,076 g, compared with 3,701 g for Adelie and 3,733 g for Chinstrap.

4 Task 2: Hypothesis testing

4.1 Two-sample t-test: body mass of Adelie vs Gentoo penguins

Hypotheses (two-sided, $\alpha = 0.05$):

- H_0: There is no difference in mean body mass between Adelie and Gentoo penguins ($\mu_{Adelie} = \mu_{Gentoo}$).
- H_1: There is a significant difference ($\mu_{Adelie} \neq \mu_{Gentoo}$).

Method. I use a *Welch* two-sample t-test. Unlike the classic Student t-test it does not assume that the two groups have equal variances, so it is the safer default. Before running it I checked the assumptions: the data in each group should be roughly normal (Shapiro-Wilk test) and the two variances should be similar (F test). A Student t-test and a Wilcoxon rank-sum test (which needs no normality assumption) are run as robustness checks.

```
adelie <- penguins$body_mass_g[penguins$species == "Adelie" &
                                 !is.na(penguins$body_mass_g)]
gentoo <- penguins$body_mass_g[penguins$species == "Gentoo" &
                                 !is.na(penguins$body_mass_g)]

# Assumption checks
shapiro_adelie <- shapiro.test(adelie)
shapiro_gentoo <- shapiro.test(gentoo)
var_res        <- var.test(adelie, gentoo)
```

Table 5: Assumption checks for the t-test.

Check	Statistic	p-value
Shapiro-Wilk, Adelie	0.981	0.0324
Shapiro-Wilk, Gentoo	0.986	0.2336
F test for equal variances	0.827	0.2691

The Adelie group shows a slight departure from normality and the Gentoo group shows no departure from normality. The F test finds no significant evidence that the variances differ. Because each group has well over 100 penguins, the t-test is robust to small departures from normality (central limit theorem), and the Welch version does not depend on equal variances.

```r
t_res <- t.test(adelie, gentoo, alternative = "two.sided",
                var.equal = FALSE, conf.level = 1 - alpha)
t_res
```

```
#>
#>  Welch Two Sample t-test
#>
#> data:  adelie and gentoo
#> t = -23.4, df = 250, p-value <2e-16
#> alternative hypothesis: true difference in means is not equal to 0
#> 95 percent confidence interval:
#>  -1491.2 -1259.5
#> sample estimates:
#> mean of x mean of y
#>    3700.7    5076.0
```

```r
t_student  <- t.test(adelie, gentoo, var.equal = TRUE)
wilcox_res <- suppressWarnings(wilcox.test(adelie, gentoo))

mean_diff <- unname(t_res$estimate[1] - t_res$estimate[2])
pooled_sd <- sqrt(((length(adelie) - 1) * var(adelie) +
                     (length(gentoo) - 1) * var(gentoo)) /
                    (length(adelie) + length(gentoo) - 2))
cohens_d  <- mean_diff / pooled_sd
```

Table 6: Results of the Adelie vs Gentoo body mass comparison.

Quantity	Value
Adelie mean (g)	3700.7
Gentoo mean (g)	5076.0
Difference, Adelie - Gentoo (g)	-1375.4
95% CI for difference (g)	[-1491.2, -1259.5]
Welch t statistic	-23.39
Degrees of freedom	249.6
p-value (Welch)	7.71e-65
p-value (Student t-test)	8.03e-68
p-value (Wilcoxon)	2.98e-42
Cohen's d (effect size)	-2.87

Result and conclusion. The Welch t-test gives $t = -23.39$ on 249.6 degrees of freedom with **p-value = 7.71e-65**. Since this is far below $\alpha = 0.05$, we **reject the null hypothesis**. The Student t-test and the Wilcoxon test lead to exactly the same conclusion, so the result does not depend on the choice of test.

In plain language. Gentoo penguins are clearly heavier than Adelie penguins. On average a Gentoo weighs 1,375 g more (about 1.4 kg), which is about 37% heavier. The 95% confidence interval says the true average gap is plausibly between 1,260 g and 1,491 g. This gap is enormous compared with the natural variation within each species (Cohen's $d \approx 2.9$; values above 0.8 are already called "large"), so it cannot reasonably be put down to chance.

4.2 One-sample proportion test: proportion of male penguins

Hypotheses (two-sided, $\alpha = 0.05$):

- H_0: The proportion of male penguins is 50% ($p = 0.5$).
- H_1: The proportion of male penguins is not 50% ($p \neq 0.5$).

Method. I use a one-sample proportion test (`prop.test`), which compares the observed share of males with 0.5. Only penguins with a recorded sex are counted. The test relies on a normal approximation, which is appropriate here because the expected counts under H_0 (about half of the sample in each group) are far larger than 5. An exact binomial test is run as a check.

```
n_known  <- sum(!is.na(penguins$sex))
n_male   <- sum(penguins$sex == "male",   na.rm = TRUE)
n_female <- sum(penguins$sex == "female", na.rm = TRUE)

prop_res  <- prop.test(x = n_male, n = n_known, p = 0.5,
                       alternative = "two.sided", correct = TRUE)
binom_res <- binom.test(x = n_male, n = n_known, p = 0.5)
prop_res
```

```
#>
#>  1-sample proportions test with continuity correction
#>
#> data:  n_male out of n_known, null probability 0.5
#> X-squared = 0.012, df = 1, p-value = 0.91
#> alternative hypothesis: true p is not equal to 0.5
#> 95 percent confidence interval:
#>  0.44957 0.55933
#> sample estimates:
#>      p
#> 0.5045
```

Table 7: Results of the test for the proportion of male penguins.

Quantity	Value
Penguins with recorded sex	333
Males	168
Females	165
Sample proportion of males	0.505
95% CI for the proportion	[0.450, 0.559]
p-value (prop.test)	0.9127
p-value (exact binomial)	0.9127

Result and conclusion. Of 333 penguins with a recorded sex, 168 are male, a sample proportion of 0.505 (50.5%). The test gives **p-value = 0.9127**, which is much larger than 0.05, so we **fail to reject the null hypothesis**. There is no evidence that the proportion of males differs from 50%.

In plain language. The sample is almost perfectly balanced: 168 males against 165 females, a difference of only 3 birds. If males and females were equally likely, a split at least this uneven would happen by chance about 91 times out of 100, so this tiny imbalance is just ordinary random variation. Note that "failing to reject" does not *prove* the true share is exactly 50%; it only means the data give no reason to doubt it.

5 Task 3: Correlation analysis

5.1 Correlation matrix

The Pearson correlation coefficient r measures the strength and direction of a *linear* relationship, from -1 (perfect negative) through 0 (none) to $+1$ (perfect positive). Penguins with missing measurements are left out (342 penguins remain).

```
cor_mat <- cor(as.data.frame(penguins)[num_vars], use = "complete.obs")
pretty_names <- c("Bill length", "Bill depth",
                  "Flipper length", "Body mass")
cor_plot_mat <- cor_mat
dimnames(cor_plot_mat) <- list(pretty_names, pretty_names)
```

Table 8: Pearson correlation matrix of the four measurements.

	Bill length	Bill depth	Flipper length	Body mass
Bill length	1.000	-0.235	0.656	0.595
Bill depth	-0.235	1.000	-0.584	-0.472
Flipper length	0.656	-0.584	1.000	0.871
Body mass	0.595	-0.472	0.871	1.000

```r
pair_list <- combn(num_vars, 2, simplify = FALSE)
cor_tests <- do.call(rbind, lapply(pair_list, function(p) {
  ct <- cor.test(penguins[[p[1]]], penguins[[p[2]]])
  data.frame(variable_1 = p[1], variable_2 = p[2],
             r = unname(ct$estimate),
             ci_low = ct$conf.int[1], ci_high = ct$conf.int[2],
             p_value = ct$p.value)
}))
```

Table 9: Correlation tests for every pair of variables (null hypothesis: the true correlation is 0).

Pair	r	95% CI	p-value
Bill length & Bill depth	-0.235	[-0.33, -0.13]	1.12e-05
Bill length & Flipper length	0.656	[0.59, 0.71]	1.74e-43
Bill length & Body mass	0.595	[0.52, 0.66]	3.81e-34
Bill depth & Flipper length	-0.584	[-0.65, -0.51]	1.23e-32
Bill depth & Body mass	-0.472	[-0.55, -0.39]	2.28e-20
Flipper length & Body mass	0.871	[0.84, 0.89]	4.37e-107

5.2 Heatmap

```r
cor_long <- as.data.frame(as.table(cor_plot_mat))
names(cor_long) <- c("Var1", "Var2", "r")

p_heat <- ggplot(cor_long, aes(x = Var1, y = Var2, fill = r)) +
  geom_tile(colour = "white", linewidth = 1) +
  geom_text(aes(label = sprintf("%.2f", r)), size = 5) +
  scale_fill_gradient2(low = "#B2182B", mid = "white", high = "#2166AC",
                       midpoint = 0, limits = c(-1, 1),
                       name = "Pearson r") +
  coord_fixed() +
  labs(title = "Correlation matrix of penguin measurements",
       x = NULL, y = NULL) +
  theme_minimal(base_size = 12) +
  theme(axis.text.x = element_text(angle = 30, hjust = 1),
        panel.grid = element_blank())
p_heat
```

5.3 Discussion of highly correlated variables

I call a pair "highly correlated" when $|r| \geq 0.70$.

- **Flipper length and body mass ($r = 0.871$) are very strongly positively correlated.** This is the only pair above the 0.70 threshold. Penguins with longer flippers are, on the whole, much heavier.

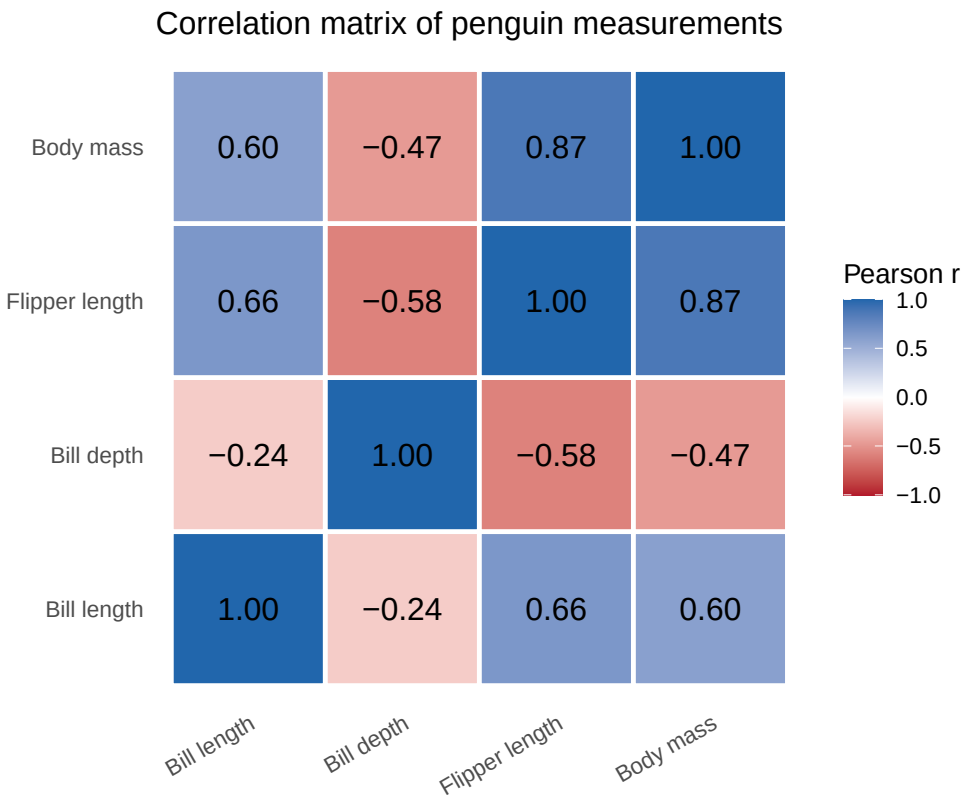


Figure 1: Heatmap of the correlation matrix. Blue = positive, red = negative; the number in each cell is the correlation coefficient.

- **Flipper length and bill length** ($r = 0.656$) and **bill length and body mass** ($r = 0.595$) are fairly strongly positively correlated: bigger birds tend to have longer bills.
- **Bill depth is negatively correlated with flipper length** ($r = -0.584$) **and with body mass** ($r = -0.472$)**.** Heavier, longer-flippered penguins tend to have *shallower* bills.
- **Bill length and bill depth are only weakly negatively correlated** ($r = -0.235$)**.**

All the correlations are statistically significant (every p-value in the table above is below 0.001), but statistical significance only says the correlation is unlikely to be exactly zero. Its *size* tells us how useful it is.

A warning about the negative correlations. The negative link between bill depth and the other measurements is partly an illusion caused by mixing three species together (a case of *Simpson's paradox*). Gentoo penguins are large with long flippers but have shallow, narrow bills, whereas the smaller Adelie and Chinstrap penguins have deeper bills. Looking *within* each species, the relationship between bill length and bill depth is actually positive:

Table 10: Correlations computed separately within each species.

Species	n	r (bill length, bill depth)	r (flipper length, body mass)
Adelie	151	0.391	0.468
Chinstrap	68	0.654	0.642
Gentoo	123	0.643	0.703

In plain language. Penguins are built "in proportion": a bird with long flippers is almost always a heavy bird, so flipper length is a very good clue to body mass. Bill shape is a different story: bill depth points in the opposite direction to size when all species are lumped together, but this is mostly because the species themselves differ in shape. Correlation never proves that one thing *causes* another.

6 Task 4: Regression modelling

6.1 Simple linear regression: body mass predicted by flipper length

```
model1 <- lm(body_mass_g ~ flipper_length_mm, data = penguins)
summary(model1)
```

```
#>
#> Call:
#> lm(formula = body_mass_g ~ flipper_length_mm, data = penguins)
#>
#> Residuals:
#>     Min      1Q  Median      3Q     Max
#> -1058.8  -259.3   -26.9   247.3  1288.7
#>
#> Coefficients:
#>                   Estimate Std. Error t value Pr(>|t|)
#> (Intercept)       -5780.83     305.81   -18.9   <2e-16 ***
```

```
#> flipper_length_mm    49.69       1.52    32.7   <2e-16 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 394 on 340 degrees of freedom
#>   (2 observations deleted due to missingness)
#> Multiple R-squared:  0.759,  Adjusted R-squared:  0.758
#> F-statistic: 1.07e+03 on 1 and 340 DF,  p-value: <2e-16
```

Regression equation (Mass = predicted body mass in grams, Flipper = flipper length in mm):

$$\widehat{\text{Mass}} = -5780.83 + 49.69 \times \text{Flipper}$$

Interpretation of the coefficients.

- **Slope (49.69).** For every extra millimetre of flipper length, the model predicts body mass to be about 49.7 g higher. A penguin with a flipper 10 mm longer is predicted to be about 497 g heavier. The 95% confidence interval for the slope is [46.70, 52.67] and the slope is highly significant ($t = 32.72$, $p < 0.001$).
- **Intercept (-5780.83).** This is the predicted body mass when flipper length is 0 mm. That is physically meaningless (no penguin has a flipper of length 0; the observed flippers run from 172 to 231 mm), so the intercept is simply the number that positions the line correctly within the observed range and should not be interpreted on its own.

Model fit. The R^2 is 0.759, meaning flipper length alone accounts for about 75.9% of the variation in body mass (adjusted $R^2 = 0.758$). The residual standard error is 394.3 g: a typical prediction is off by about this much. The overall F-test is highly significant ($F = 1070.7$, $p < 0.001$).

Table 11: Simple regression: coefficient estimates.

Term	Estimate	Std. error	t value	p-value	95% CI
Intercept	-5780.83	305.81	-18.90	5.59e-55	[-6382.36, -5179.30]
Flipper length (mm)	49.69	1.52	32.72	4.37e-107	[46.70, 52.67]

6.2 Multiple linear regression

```
model2 <- lm(body_mass_g ~ flipper_length_mm + bill_length_mm +
               bill_depth_mm, data = penguins)
summary(model2)
```

```
#>
#> Call:
#> lm(formula = body_mass_g ~ flipper_length_mm + bill_length_mm +
#>     bill_depth_mm, data = penguins)
#>
```

```
#> Residuals:
#>     Min      1Q  Median      3Q     Max
#> -1054.9  -290.3   -21.9   239.0  1276.6
#>
#> Coefficients:
#>                    Estimate Std. Error t value Pr(>|t|)
#> (Intercept)       -6424.76     561.47  -11.44   <2e-16 ***
#> flipper_length_mm    50.27       2.48   20.29   <2e-16 ***
#> bill_length_mm        4.16       5.33    0.78     0.44
#> bill_depth_mm        20.05      13.69    1.46     0.14
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 393 on 338 degrees of freedom
#>   (2 observations deleted due to missingness)
#> Multiple R-squared:  0.761,  Adjusted R-squared:  0.759
#> F-statistic:  360 on 3 and 338 DF,  p-value: <2e-16
```

Regression equation (Mass in grams; Flipper, BillLen and BillDepth are flipper length, bill length and bill depth in mm):

$$\widehat{\text{Mass}} = -6424.76 + 50.27 \times \text{Flipper} + 4.16 \times \text{BillLen} + 20.05 \times \text{BillDepth}$$

Table 12: Multiple regression: coefficient estimates.

Term	Estimate	Std. error	t value	p-value	95% CI
Intercept	-6424.76	561.47	-11.44	7.58e-26	[-7529.18, -5320.35]
Flipper length (mm)	50.27	2.48	20.29	1.93e-60	[45.40, 55.14]
Bill length (mm)	4.16	5.33	0.78	0.4354	[-6.32, 14.64]
Bill depth (mm)	20.05	13.69	1.46	0.1441	[-6.89, 46.99]

Significance of each predictor (each coefficient is the effect of that variable *holding the other two constant*):

- **Flipper length** (50.27 g per mm, $p < 0.001$) is highly significant. It is the real driver of the model.
- **Bill length** (4.16 g per mm, $p = 0.4354$) is **not** significant once flipper length is known: it adds no useful information.
- **Bill depth** (20.05 g per mm, $p = 0.1441$) is **not** significant for the same reason.

```r
predictors <- c("flipper_length_mm", "bill_length_mm", "bill_depth_mm")
vif_values <- sapply(predictors, function(v) {
  others <- setdiff(predictors, v)
  r2 <- summary(lm(reformulate(others, response = v),
                   data = penguins))$r.squared
  1 / (1 - r2)
})
round(vif_values, 2)
```

```
#> flipper_length_mm    bill_length_mm     bill_depth_mm
#>              2.67              1.87              1.61
```

The variance inflation factors are all below 5, so multicollinearity (the predictors overlapping too much with each other) is mild and the coefficient estimates are stable.

6.3 Comparing the two models

```
model_anova <- anova(model1, model2)
model_anova
```

```
#> Analysis of Variance Table
#>
#> Model 1: body_mass_g ~ flipper_length_mm
#> Model 2: body_mass_g ~ flipper_length_mm + bill_length_mm + bill_depth_mm
#>   Res.Df      RSS Df Sum of Sq    F Pr(>F)
#> 1    340 52854796
#> 2    338 52311359  2    543437 1.76   0.17
```

```
model_comparison <- data.frame(
  Model = c("Simple (flipper length)",
            "Multiple (flipper + bill length + bill depth)"),
  Predictors = c(1, 3),
  R2     = c(s1$r.squared,     s2$r.squared),
  AdjR2  = c(s1$adj.r.squared, s2$adj.r.squared),
  RSE    = c(s1$sigma,         s2$sigma),
  AIC    = c(AIC(model1),      AIC(model2)),
  BIC    = c(BIC(model1),      BIC(model2))
)
```

Table 13: Comparison of the simple and multiple regression models (lower AIC, BIC and RSE are better; higher R2 is better).

Model	Predictors	R2	Adj. R2	RSE (g)	AIC	BIC
Simple (flipper length)	1	0.7590	0.7583	394.3	5062.9	5074.4
Multiple (flipper + bill length + bill depth)	3	0.7615	0.7594	393.4	5063.3	5082.5

Comparison. Adding the two bill measurements raises R^2 only from 0.759 to 0.761 (an increase of 0.25 percentage points) and adjusted R^2 from 0.758 to 0.759. (Plain R^2 can never go down when predictors are added, so adjusted R^2 is the fairer comparison.) The residual standard error barely moves (394.3 g vs 393.4 g). The formal F-test comparing the models gives p = 0.1744, so the extra predictors do **not improve the model significantly**. The information criteria agree: AIC is 5062.9 for the simple model and 5063.3 for the multiple model, and BIC (5074.4 vs 5082.5) clearly prefers the simpler model because it penalises unnecessary predictors more heavily.

In plain language. Knowing a penguin's flipper length already tells us almost everything the bill measurements could add about its weight. The simple model is just as accurate and much easier to explain, so it is the better choice (the principle of parsimony: prefer the simplest model that works).

6.4 Checking the regression assumptions

```
par(mfrow = c(2, 2))
plot(model2)
```

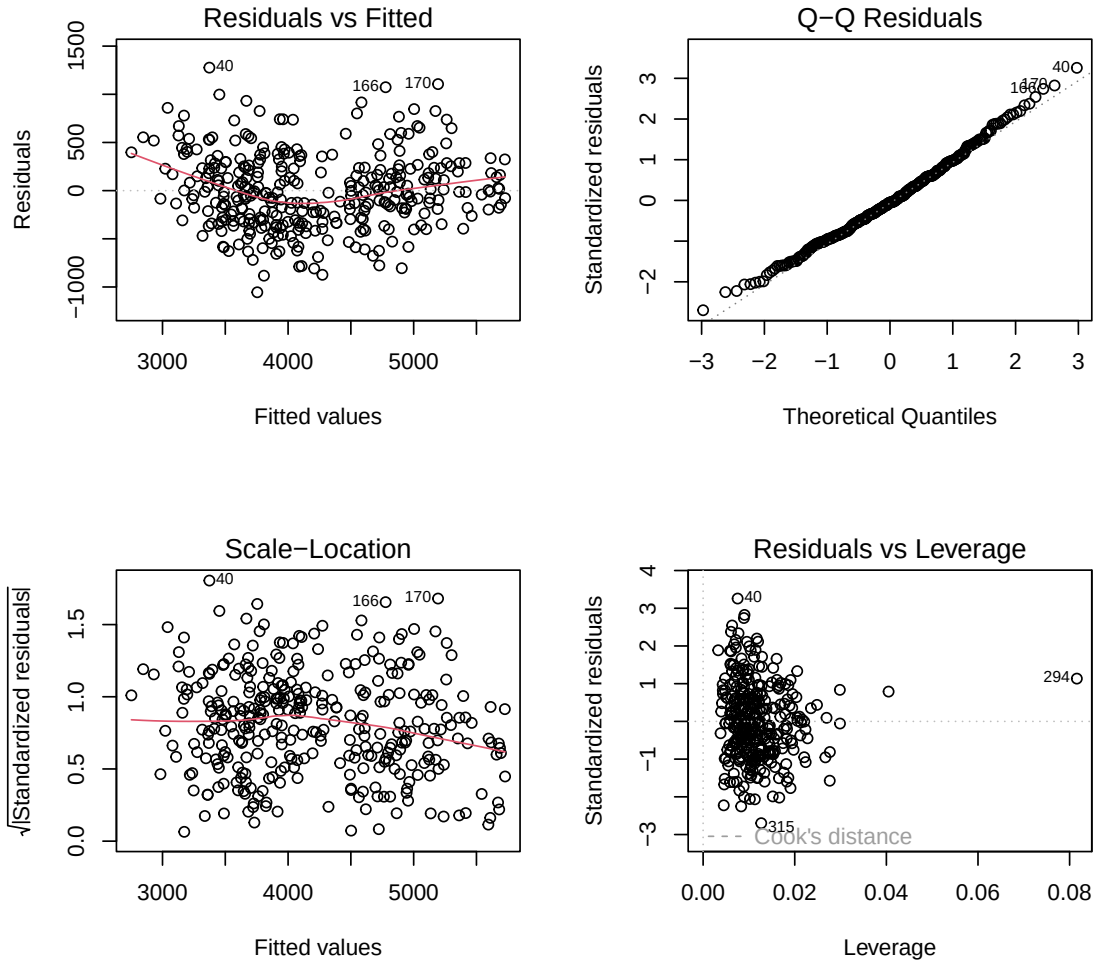


Figure 2: Diagnostic plots for the multiple regression model.

```
par(mfrow = c(1, 1))
```

The four panels check, in turn, whether the relationship is linear (residuals vs fitted values should show no pattern), whether the residuals are roughly normal (points near the line in the Q-Q plot), whether the spread of the residuals is constant (scale-location plot) and whether any single penguin has an outsized influence (residuals vs leverage). Each of these is also backed by a numerical check:

```r
mf <- model.frame(model2)

# 1. Normality of the residuals (Shapiro-Wilk)
shapiro_resid <- shapiro.test(residuals(model2))

# 2. Constant spread: Breusch-Pagan (studentised) test, calculated by hand
aux <- lm(residuals(model2)^2 ~ flipper_length_mm + bill_length_mm +
            bill_depth_mm, data = mf)
bp_stat <- nobs(model2) * summary(aux)$r.squared
bp_p    <- pchisq(bp_stat, df = 3, lower.tail = FALSE)

# 3. Influence: Cook's distance and large standardised residuals
max_cook   <- max(cooks.distance(model2))
n_outliers <- sum(abs(rstandard(model2)) > 3)

# 4. Linearity: RESET test (does adding the squared fitted values help?)
mf$fit2   <- fitted(model2)^2
reset_fit <- lm(body_mass_g ~ flipper_length_mm + bill_length_mm +
                  bill_depth_mm + fit2, data = mf)
reset_p   <- anova(model2, reset_fit)$"Pr(>F)"[2]
```

- **Normality of residuals.** The Shapiro-Wilk test gives p = 0.1640, so the normality assumption is satisfied.
- **Constant spread.** The Breusch-Pagan test gives p = 0.3115: there is no evidence that the spread of the errors changes with the predictors.
- **Influential points.** The largest Cook's distance is 0.029 (values above 0.5 would be a concern), so no single penguin drives the results. The number of penguins with a standardised residual beyond ±3 is 1; a few such unusually light or heavy birds are expected in a sample of this size.
- **Linearity.** The RESET test gives p = 2.29e-06: the straight-line model leaves a small amount of structure unexplained. The most likely reason is the species clusters visible in the scatter plot below, which a single line cannot capture (see the limitations section).

7 Task 5: Visualization

7.1 Scatter plot of body mass vs flipper length with the regression line

```r
pen_fl_bm <- penguins[!is.na(penguins$flipper_length_mm) &
                        !is.na(penguins$body_mass_g), ]
eq_label <- sprintf(
  "Body mass = %.1f + %.2f x Flipper length\nR-squared = %.3f",
  b1[1], b1[2], s1$r.squared)

p_scatter <- ggplot(pen_fl_bm, aes(flipper_length_mm, body_mass_g)) +
  geom_point(aes(colour = species), alpha = 0.75, size = 2) +
  geom_smooth(method = "lm", formula = y ~ x, se = TRUE,
              colour = "black", linewidth = 0.9) +
```

```
  annotate("text", x = min(pen_fl_bm$flipper_length_mm) + 1,
           y = max(pen_fl_bm$body_mass_g), label = eq_label,
           hjust = 0, vjust = 1, size = 3.8) +
  scale_colour_manual(values = species_cols, name = "Species") +
  labs(title = "Body mass vs flipper length",
       x = "Flipper length (mm)", y = "Body mass (g)") +
  theme_minimal(base_size = 12)
p_scatter
```

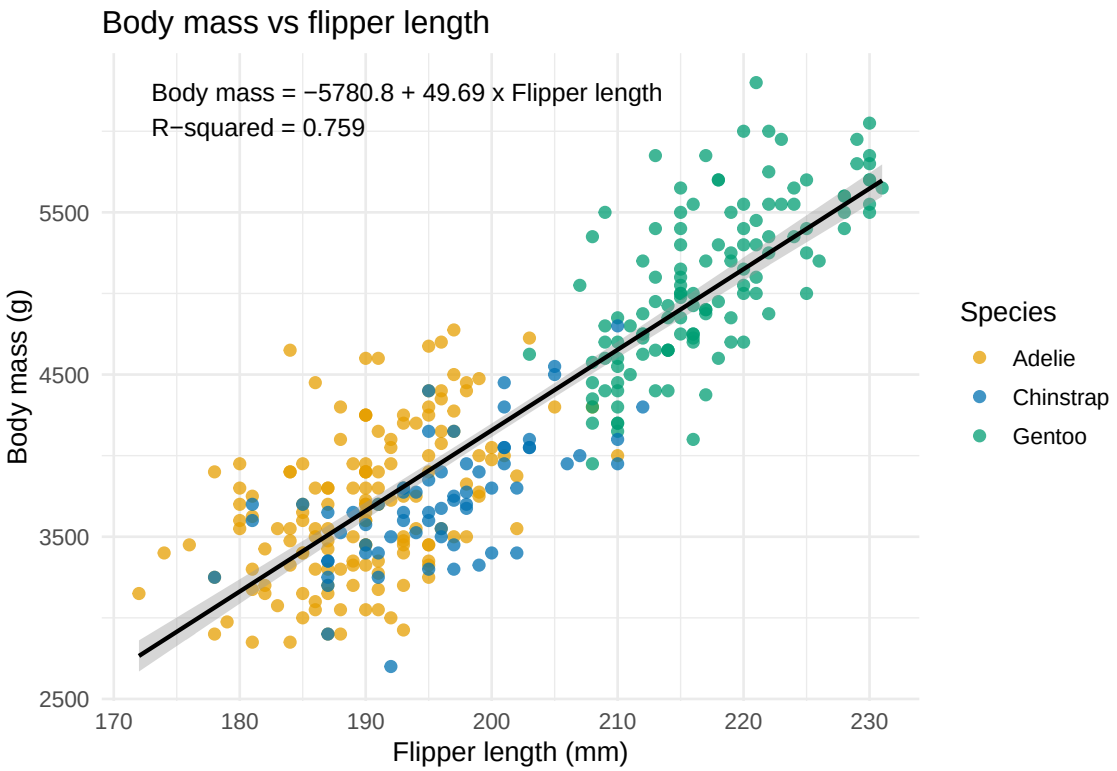


Figure 3: Body mass against flipper length, with the least-squares regression line and its 95% confidence band.

The points rise steadily from bottom left to top right and cluster tightly around the line, which is what the high correlation ($r = 0.87$) and R^2 of 0.76 predicted. Colouring the points by species adds an important detail: the Gentoo penguins (green) form a separate cluster at the top right, whereas Adelie (orange) and Chinstrap (blue) overlap at the lower left. A good part of the overall trend therefore reflects the difference *between* species as well as the difference between individual birds.

7.2 Boxplot of body mass across species

```
pen_bm_sp <- penguins[!is.na(penguins$body_mass_g), ]

p_box <- ggplot(pen_bm_sp, aes(species, body_mass_g, fill = species)) +
  geom_boxplot(alpha = 0.8, outlier.shape = NA, width = 0.55) +
```

```
  geom_jitter(width = 0.12, alpha = 0.35, size = 1.3) +
  stat_summary(fun = mean, geom = "point", shape = 23, size = 3,
               fill = "white") +
  scale_fill_manual(values = species_cols) +
  guides(fill = "none") +
  labs(title = "Distribution of body mass by species",
       x = "Species", y = "Body mass (g)") +
  theme_minimal(base_size = 12)
p_box
```

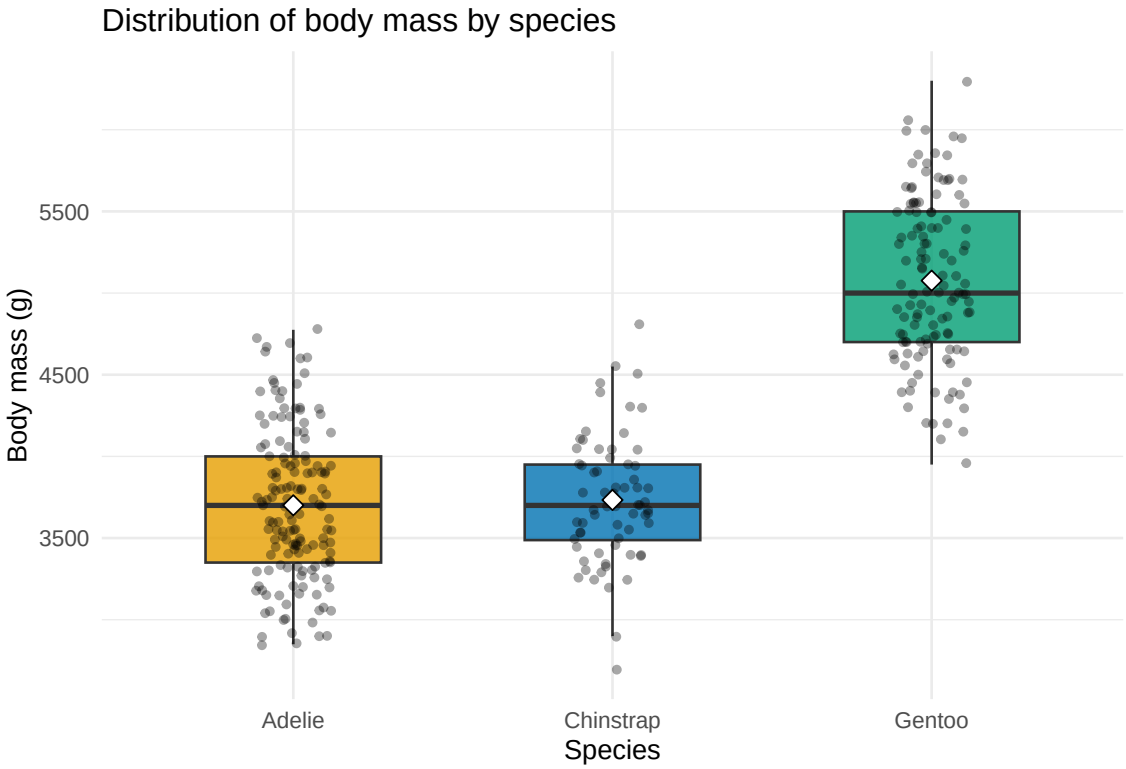


Figure 4: Distribution of body mass by species (diamond = group mean).

Gentoo penguins (median 5,000 g) sit far above the other two species, and their whole box lies above the Adelie and Chinstrap boxes: there is almost no overlap between the typical Gentoo and the typical Adelie, which matches the t-test result. Adelie (median 3,700 g) and Chinstrap (median 3,700 g) have very similar medians. The Adelie and Gentoo boxes have broadly similar heights, so their spread of weights is similar, which fits the F test result and supports the t-test. The grey dots (the individual penguins) show the data behind each box.

7.3 Histogram of body mass with a normal curve

```
bm <- penguins$body_mass_g[!is.na(penguins$body_mass_g)]

p_hist <- ggplot(data.frame(body_mass_g = bm), aes(x = body_mass_g)) +
```

```
  geom_histogram(aes(y = after_stat(density)), bins = 25,
                 fill = "#8DA0CB", colour = "white") +
  stat_function(fun = dnorm, args = list(mean = mean(bm), sd = sd(bm)),
                colour = "#B2182B", linewidth = 1) +
  labs(title = "Histogram of body mass with fitted normal curve",
       x = "Body mass (g)", y = "Density") +
  theme_minimal(base_size = 12)
p_hist
```

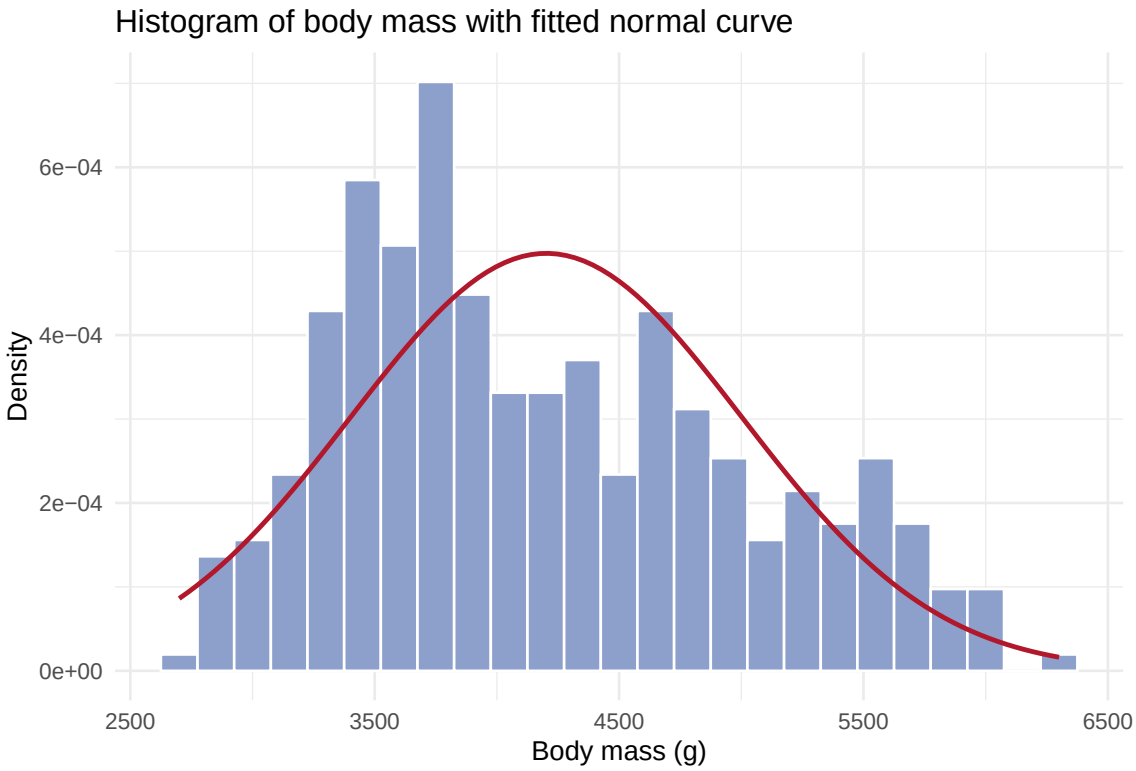


Figure 5: Histogram of body mass with a normal curve using the sample mean and standard deviation.

```
skew_bm    <- mean((bm - mean(bm))^3) / sd(bm)^3
shapiro_bm <- shapiro.test(bm)
```

The red curve is the bell shape we would see if body mass were perfectly normal with the same mean (4,202 g) and standard deviation (802 g). The actual histogram departs from it: it is **right-skewed** (skewness = 0.47), with a long tail of heavy penguins and a second hump at the heavy end. The Shapiro-Wilk test confirms that body mass is not normally distributed overall (p = 3.68e-08). The reason is that the data are a *mixture*: the lighter Adelie and Chinstrap penguins form the main hump and the heavier Gentoo penguins form the second. Within each species the distribution is much closer to normal, which is why the group comparisons in Task 2 are still valid.

8 Summary of findings in plain language

- **The data are in good shape.** Only 11 of 344 penguins have any missing information, so almost nothing was lost.
- **Gentoo penguins are much heavier than Adelie penguins**, by about 1,375 g on average. The difference is far too large to be chance (p = 7.71e-65).
- **Males and females are about equally common** in the sample (168 vs 165); there is no evidence of an imbalance (p = 0.9127).
- **Flipper length is the best single clue to body mass.** The two are very strongly correlated (r = 0.87), and each extra millimetre of flipper goes with about 49.7 g of extra weight.
- **A simple model is good enough.** Flipper length alone explains 76% of the variation in body mass; adding bill length and bill depth adds almost nothing (0.759 to 0.761) and is not statistically worthwhile.
- **Species matters.** Much of what looks like a relationship between measurements is partly a difference between species (Gentoo are bigger in almost every respect except bill depth).

9 Assumptions and limitations

9.1 Assumptions behind the methods

Method	Assumption	How it was checked	Outcome
t-test	Independent observations	Not testable from the data	Assumed (see limitations)
t-test	Roughly normal data in each group	Shapiro-Wilk test, large samples	Acceptable (CLT; Wilcoxon agrees)
t-test	Equal variances	F test; Welch version used anyway	Not needed with Welch
Proportion test	Independent trials; large enough sample	Expected counts far above 5	Satisfied
Correlation	Linear relationship, no extreme outliers	Scatter plots, boxplot	Reasonable
Regression	Linearity, constant spread, normal residuals, independence	Diagnostic plots, Shapiro-Wilk	Broadly satisfied
Multiple regression	Predictors not too collinear	Variance inflation factors	VIF below 5 for all

9.2 Limitations

1. **Species is a hidden driver.** Pooling the three species inflates the flipper-body mass relationship and reverses some bill correlations (Simpson's paradox). As a check, adding species to the regression (`body_mass_g ~ flipper_length_mm + species`) cuts the flipper slope from 49.7 to 40.7 g per mm and raises R^2 to 0.783. Within a species, flipper length is a somewhat weaker predictor than the pooled result suggests.

2. **Observational data, no causation.** The penguins were observed, not experimentally manipulated, so the correlations and regression slopes describe association only. Longer flippers do not *cause* a heavier body.
3. **Independence is assumed.** The dataset does not say whether the same bird could have been measured more than once across the 3 observation years, or whether birds from the same nest are related. Either would break the independence assumption slightly.
4. **Limited scope.** The sample covers three islands over a few years, so the results describe these colonies and should not be generalised to all penguins. The male/female result concerns the *sample*, which may not reflect how the birds were selected.
5. **Missing data.** Dropping incomplete rows is reasonable here because so few are missing, but it assumes the missing penguins are not systematically different from the others.
6. **Statistical vs practical significance.** With 300+ observations even tiny effects can be "significant". That is why effect sizes (Cohen's d, R^2, confidence intervals) are reported alongside p-values.
7. **Multiple tests.** Several tests were run at $\alpha = 0.05$ without adjustment. The main conclusions (p-values far below or far above 0.05) are not sensitive to this.
8. **Extrapolation.** The regression equations are only valid within the observed flipper lengths (172 to 231 mm); predictions outside that range (including the intercept at 0 mm) are not reliable.

10 Conclusion

Within this dataset, species is the strongest organising factor. Gentoo penguins are significantly and substantially heavier than Adelie penguins, the sexes are evenly balanced, and body mass is predicted well by flipper length alone ($R^2 = 0.76$), with no real gain from adding bill length and depth. The main caution is that species differences drive part of every relationship, so any use of these models should account for species.

11 Appendix: reproducibility

Table 14: Software versions (R version 4.6.1 (2026-06-24 ucrt)).

Package	Version
palmerpenguins	0.1.1
ggplot2	4.0.3
dplyr	1.2.1
knitr	1.52
rmarkdown	2.32

To reproduce this report, place `penguins_analysis.R` and `DTS204_Lab_Report.Rmd` in the same folder, then run `rmarkdown::render("DTS204_Lab_Report.Rmd")`. To reproduce only the numerical results and the figures, run `source("penguins_analysis.R")`.